INCH-POUND

MIL-P-14895E (AR)
AMENDMENT 5
11 April 1991

SUPERSEDING
AMENDMENT 4
30 November 1988

MILITARY SPECIFICATION

PROJECTILE, 8 INCH, M509A1
METAL PARTS ASSEMBLY

This Amendment forms a part of Military Specification MIL-P-14895E (AR), dated 25 April 1983, and is approved for us within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1.1 Specifications, Military: Add "MIL-A-8625 - Anodic Coatings for Aluminum and Aluminum Alloys".

PAGE 2

2.1.2 Inspection Equipment Drawings: Delete "9278669" and substitute "9280361".

Add: "9280362 - Envelope Drawing Eddy Current Equipment for M509 Base".

PAGE 3

3.2.3.1 Delete in its entirety.

PAGE 4

* Add the following new paragraphs:

"3.3.1.1 Minimum melting of steel. Upon removal of the rotating band material, the steel band seats shall demonstrate visual evidence of minimum melting due to copper adhesion during application of the rotating band. Visual comparison standards will be provided by the Government.

3.3.1.2 Subsurface defects. There shall be no evidence of either intergranular penetration of copper or weld start-stop cratering when the body band seats are turned down to a 7.76 plus or minus 0.010 inch diameter and tested as specified in 4.5.3.1."

3.3.2.1 Delete in its entirety.

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PAGE 5

3.6.1 Delete in its entirety and substitute the following:

"3.6.1 Microstructure. The microstructure of the ogive and base shall comply with the microstructure depicted in Figures 1A, 1B or 2. Two examples of acceptable microstructure are shown in Figures 1A and 1B. In Figure 1A, the insoluble phase is well broken up, randomly distributed within a matrix which is essentially unrecrystallized. In Figure 1B, the insoluble phase is well broken up and randomly distributed within a matrix which is partially recrystallized. An example of a marginally acceptable microstructure is shown in Figure 2. In this figure the insoluble second phase exhibits some discontinuity and appears to be elongated. An example of an unacceptable microstructure is shown in Figure 3. The insoluble phase is distributed in a more continuous form (network) within a matrix which shows a large degree of microsegration (or coring). The microstructure in Figure 3 must be avoided because it is detrimental to the integrity of the material."

PAGE 6

Add the following paragraphs:

"3.14 Protective finish. Details of the proposed procedure for zinc phosphate coating and anodic coating (with duplex seal) to be used by the contractor shall be submitted in writing to the Government technical agency and written approval received prior to the initiation of production. The proposed procedure shall include materials, process and equipment to be used by the contractor, detailed method of control including limits for time, temperature, concentration and all other information pertinent to the processes. Approval of process, materials, or equipment implies no guarantee of acceptance of the results obtained in use.

3.14.1 Ogive and base. First article ogives and bases shall withstand the salt spray test prescribed in paragraph 3.10.1.2 of MIL-A-8625. The production lot components shall also comply with 3.10.1.2 except the test duration shall be 72 hours and shall show no more than 3 corrosion spots or pits on the significant surface."

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PAGE 7

4.3.1 Delete in its entirety the part entitled "Obturator (Dwg. 9287983)."

PAGE 9

4.3.2 Table I. Delete in its entirety the part entitled "Obturator (Dwg. 9287983)."

PAGE 16

4.4.2.2 Delete in its entirety.

PAGE 17

4.4.2.3 Delete in its entirety and substitute new page 17 as included.

Delete Major 107 and substitute the following:

"Length from front face to rear end of threads	-	0.40%	3.1	Gage".

PAGE 18

4.4.2.3 Delete in its entirety and substitute new page 18 as included.

Delete Minor 204 and substitute the following:

"Length from front face to end of thread relief groove	-	0.65%	3.1	Gage".

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.3	Base Before Anodizing				9343032
					NEXT HIGHER ASSEMBLY
					10551896
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Metal defective		100%	3.1	4.5.9
Major					
101	Forward runout on boat-tail with pitch diameter of thread and shoulder		0.40%	3.1	Gage
102	Rear runout on boat-tail with pitch diameter of thread and shoulder		0.40%	3.1	Gage
103	Straightness of a diametral element of shoulder		0.40%	3.1	Gage
104	Length from shoulder to forward basic diameter on boat-tail		0.40%	3.1	Gage
105	Length from shoulder to rear basic diameter on boat-tail		0.40%	3.1	Gage
106	Length from front face to shoulder		0.40%	3.1	Gage
107	Length from front face to end of thread relief groove		0.40%	3.1	Gage
108	Depth of wrench hole (2 places)		0.40%	3.1	Gage
109	Width of o-ring groove		0.40%	3.1	Gage
110	Thickness through base web		0.40%	3.1	Gage
111	Diameter of flat		0.40%	3.1	Gage
112	Diameter of o-ring groove		0.40%	3.1	Gage

NOTES:

4 (17)

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9343032
4.4.2.3	Base Before Anodizing				NEXT HIGHER ASSEMBLY 10551896
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
113	True position of flat to thread pitch diameter		0.40%	3.1	Gage
114	Pitch diameter of thread		0.40%	3.1	Gage
115	Major diameter of thread		0.40%	3.1	Gage
116	Yield strength and percent elongation	*		3.6.2	4.5.1.5
117	Microstructure	*		3.1	4.5.6
118	Minor diameter of thread, max.	**	0.40%	3.1	Gage
Minor					
201	Depth from cavity basic diameter to bottom of cavity		0.65%	3.1	Gage
202	Length from rear surface to shoulder		0.65%	3.1	Gage
203	Diameter of thread relief groove		0.65%	3.1	Gage
204	Width of thread relief groove		0.65%	3.1	Gage
205	Presence and location of wrench holes		0.65%	3.1	Visual/Func. Gage
206	Radii or chamfers missing		0.65%	3.1	Visual
207	Surface finish improper		0.65%	3.1	Visual
208	Burr		0.65%	3.1	Visual
209	Evidence of poor workmanship		0.65%	3.13	Visual

NOTES: * Sample size in accordance with inspection method paragraph.
** Applies only when threads are anodized.

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PAGE 19

4.4.2.4, Major 101: Delete in its entirety.

PAGE 20

4.4.2.5, Title: Delete "(Prior to final machining of boat-tail)".

Under Special Category:

* Special a. Band seat metal defective change "3.4.2" to "3.3.1/3.4.2".

Delete Special b in its entirety and add the following:

"c	Base threads damaged or not full	100%	3.1	Visual"

Critical: Delete "None defined" and replace with the following:

"1	Metal defective	100%	3.4.1	4.5.3.2"

Majors 104 and 105: Delete in their entirety.

PAGE 22

4.4.2.5, Major 127: Delete "undercut".

Add "***" to 134, 135 and 136. Add note "***" alternate inspections, see paragraph 4.5.13.

PAGE 23

4.4.2.5, Major 137: Add "***".

Add note "***" alternate inspection, see paragraph 4.5.13.

Major 138: Delete in its entirety.

Major 140: Delete "undercut".

Major 147: Delete in its entirety.

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Add the following Major defects:

"148	Depth from rear face to basic diameter forward of base thread	0.40%	3.1	Gage
149	Min. depth of full form base thread	0.40%	3.1	Gage
150	True position of rear counterbore to base thread	0.40%	3.1	Gage
151	Straightness of a diametral element of the rear surface	0.40%	3.1	Gage
152	Pitch diameter of base thread	0.40%	3.1	Gage
153	Minor diameter of base thread	0.40%	3.1	Gage
154	Runout of base thread pitch diameter with bourrelets and rear surface	0.40%	3.1	Gage
155	Runout of boat-tail with bourrelets and rear surface	0.40%	3.1	Gage
156	Location of index mark with keyway	0.40%	3.1	Gage
157	Length from rear end to outside basic diameter on boat-tail	0.40%	3.1	Gage
158	Diameter of rear counterbore	0.40%	3.1	Gage"

PAGE 24

4.4.2.5, Minor Category: Add the following:

"212	Location of marking	0.65%	3.1	Gage
213	Marking missing, incorrect or illegible	0.65%	3.1	Visual
214	Protective coating incomplete or damaged	0.65%	3.1	Visual"

PAGE 25

4.4.2.6: Delete in its entirety.

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PAGE 27

4.4.2.8 Special, Add:

"b	Lifting plug or gasket missing	100%	3.1	Visual"

Delete Major 108.

Major defect 105: Delete in its entirety.

PAGE 28

4.4.2.8 Minor defect 203: Delete in its entirety.

Add Minor:

"210	VCI paper missing	0.65%	3.1	Visual"

Add new paragraph 4.4.2.9 included as page 28a.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9229038
4.4.2.9	Pallet				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a.	Cover support missing (6 places)		100%	3.1	Visual
Major	None defined				
Minor					
201	Pallet damaged (unserviceable)		0.65%	3.1	Visual
202	Improperly packed		0.65%	3.1	Visual
203	Strapping missing, loose or broken		0.65%	3.1	Visual/Manual
204	Marking missing, incorrect or illegible		0.65%	3.1	Visual
205	Evidence of poor workmanship		0.65%	3.13	Visual

NOTES:

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Add new paragraph 4.4.2.10 included.

4.4.3.3 Delete in its entirety.

4.4.3.4 Delete in its entirety.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9343032
4.4.2.10	Base After Anodizing				NEXT HIGHER ASSEMBLY 10551896
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a.	Threads damaged or not full		100%	3.1	Visual
b.	Uncoated aluminum in non-allowed area		100%	3.1	Visual
Major					
101	Pitch diameter of thread	**	0.40%	3.1	Gage
102	Major diameter of thread	**	0.40%	3.1	Gage
103	Minor diameter of thread	**	0.40%	3.1	Gage
104	Protective finish	*	-	3.1	Gage
Minor					
201	Coating incomplete or damaged		0.65%	3.1	Visual
202	Evidence of poor workmanship		0.65%	3.13	Visual

NOTES: * Sample size in accordance with inspection method paragraph.
** Applies only when threads are anodized.

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* 4.5.3.1 Delete in its entirety and substitute the following:

"4.5.3.1 Inspection of Band Seat.

4.5.3.1.1 Welder qualification sampling. Initially the first five banded bodies after the welded overlay band has been applied, for each welding machine in operation, shall be subjected to the tests outlined below. Reduced sampling of one in twenty banded bodies may be implemented after five consecutive banded bodies per each welding machine in operation have been found acceptable. These bodies shall have the band removed by machining and chemically etching without damaging the band seat metal (steel) (see 6.8). These bodies will be subjected to the following tests:

a. Visual examination for compliance with 3.3.1.1 minimum melt requirement.

b. Visual examination for compliance with 3.4.2 metal defects, using the technique described in 4.5.3.1.3.

c. Turn down band seats to the diameter specified in 3.3.1.2 and again visually inspect in accordance with 4.5.3.1.3. Any defect indication per 3.3.1.2 at this depth is considered to be a reject.

d. Should any sample for any of the above tests fail to comply with the minimum acceptance requirements, the quantity of bodies represented by that sample shall be rejected.

4.5.3.1.2 Post Qualification Sampling.

4.5.3.1.2.1 Etched bandseat examination. Reduced sampling of one in each 8 hours production or fraction thereof per welding machine in operation may be implemented after five consecutive samples of one in twenty banded bodies per each welding machine in operation have been found acceptable (see 6.9). One body per welder will be randomly selected and inspected according to the following criteria:

a. Visual examination for compliance with 3.3.1.1, minimum melting. The body selected for this test need not be the test sample having the lower rotating band iron content when tested in accordance with 4.5.4.

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b. Visual inspection for compliance with 3.4.2, metal defects, using the technique described in 4.5.3.1.3. If the sample fails to comply with 3.4.2 then the band seats will be turned down to the diameter specified in 3.3.1.2 and again visually inspected in accordance with 4.5.3.1.3.

c. Should any sample fail to comply with the minimum acceptance requirements, the quantity of bodies represented by that sample shall be rejected.

d. Two consecutive failures when tested for a, or two consecutive failures when tested for b, for any welding machine will mandate the return to the original welder qualification sample of paragraph 4.5.3.1.1. However, reduced sampling of one in each 8 hours production or fraction thereof may be implemented after the first five banded bodies have been found acceptable.

4.5.3.1.2.2 Sub-surface defect examination. The body having the highest acceptable iron content rotating band when tested per paragraph 4.5.4 during this 8 hour post qualification sampling shall be tested for compliance with 3.3.1.2. This body will have its band seats turned down to the diameter specified in paragraph 3.3.1.2 and visually inspected per paragraph 4.5.3.1.3. Failure of the body to comply with the specified requirements will require rejection of all bodies having bands welded by the same welder during the same sampling period. Failure of any body will also require that the body exhibiting the highest iron content from each welder will be similarly tested. Failure of any body will result in rejection of the quantity from each discrete welder represented by the sample."

PAGE 34

4.5.4, line 15: Delete "dissolved and the analysis made from the individual solutions" and substitute "analyzed by any method approved by the technical agency".

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PAGE 35

4.5.6: Delete and substitute the following:

"4.5.6 Microstructure examination (ogive and base). One ogive and one base forging from each heat treatment lot shall be selected for microstructure examination. The metallographic specimen shall be etched with Keller's Reagent and all areas examined under a microscope/metallograph at a minimum magnification of 250 diameters. Any evidence of the insoluble phase generally occurring in a semi-continuous network form shall be cause for rejection of the lot represented by the sample. Microstructural preparation and examination shall be performed by a metallographer or metallurgical engineer familiar with aluminum alloys."

PAGE 36

4.5.8.1 Ogive salt spray test. The significant surface may be marked by tracing the outline of a 5 inch by 6 inch sheet of paper onto exterior of the ogive.

4.5.9, line 6: Delete "9278669" and substitute "9280361 and 9280362".

Add the following new paragraph:

"4.5.9.1 Operation. As a minimum the adequacy of the inspection system shall be checked by the use of standards, 9280361, at the start and finish of each inspection shift or change of operator, as well as at two hour intervals during inspection. All channels shall properly indicate a reject signal when a defect is encountered. If the inspection process is determined to be inadequate, the condition shall be corrected and all items inspected since the last acceptable inspection check shall be reinspected. A log book shall be maintained by the operator of each system which as a minimum will document date and time, adjustments made, number of times inspected, number of rejects, changing of sensors, instrumentation repairs, reasons for malfunctions, if any, and changing of qualified operators. This log book shall be made available when requested for Government review.

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PAGE 37

4.5.10.2 Delete line 5: "A zone 1 propellant...shall be used."

Add line 5:

"A M1 propellant charge for zone 1 shall be used."

4.5.10.3 Delete line 4: "A zone 9 propellant...shall be used."

Add line 4:

""A M188A1 propellant charge for zone 9 shall be used."

PAGE 38

4.5.11 Delete in its entirety.

4.5.12 Delete in its entirety.

4.5.13 Add the following paragraph:

4.5.13 Alternate inspection of dovetail grooves. 100% inspection required. The width of the dovetail grooves (major defects 134, 135, 136 and 137, paragraph 4.4.2.5) may be gaged with a functional check in lieu of inspection of the actual characteristics. Each keyway shall accept and retain both a maximum metal key and a minimum metal key (part numbers 9210134 and 9210135) throughout its entire full form length."

Delete paragraphs 5.1.1, 5.1.2, 5.1.3.

Delete paragraph 5.2.2 and substitute:

"5.2.2 Bases shall be packaged and packed in accordance with contractor design packaging. Packaging will comply with MIL-STD-3169, Level 3 for bulk packaging."

5.3, line 2: Delete "8837839" and substitute "9229038".

line 3: Add "Level B" after MIL-STD-1169.

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PAGE 39

6.4 Delete "4.5.12" from paragraph column for Quality Inspection Test Demonstration & Evaluation Report.

Delete "4.5.11" from paragraph column for Test and Demonstration Report, DI-T-1906.

PAGE 43

Delete title, "Figure 2 Unacceptable Microstructures (250X)"

Figure 2a, change "a" for "Figure 2 marginally acceptable microstructure (250X)".

Figure 2b, change "b" to "Figure 3 unacceptable microstructure (250X)".

PAGE 45

Appendix A: Delete in its entirety.

The margins of this amendment are marked with an asterisk or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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